

# Asandile Nkala

## Software Engineer

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### SUMMARY

I'm a passionate **Software Engineer** specializing in **Software Development**. With a keen interest in solving complex problems and dedication to continuous learning. I thrive in creating efficient and innovative software solutions. My journey in Technology has equipped me with a robust skill set, including proficiency in various Programming Languages and Technologies. I am motivated by the ever-evolving landscape of Software Development and I am committed to contributing to impactful Projects that drive progress and innovation.

### TECHNICAL SKILLS

**Java, Python, C#, Mongo, Express, React, Node, Relational Database, SQL, Docker, Postman, API's, Agile Methodologies, Git & GitHub, Documentation, Linux Command, Google Cloud, Supabase & AWS .**

### Soft SKILLS

**Adaptability, Communication, Problem-Solving, Teamwork and Collaboration, Time Management, Empathy.**

### PERSONAL PROJECTS

[More Projects on

GitHub]

#### **Eduvate : [Project in progress]**

- Eduvate is a comprehensive educational technology platform designed to enhance learning through AI-powered personalized assistance and interactive assessments. It provides students with intelligent tutoring, automated question generation, and real-time feedback across multiple subjects for grades k-12. The platform features AI-driven capabilities including content summarization, translation, progress analysis, and conversational learning support. Built with Spring Boot and integrating Spring AI for Gemini-powered features, Eduvate creates an adaptive learning environment that caters to individual student needs and learning paces.
- **Tools:** Java, Supabase, React.

- **GitHub Repo:** Both frontend and backend repos are private

### **Poll Application with User Authentication: [HyperionDev]**

- This project extends the basic functionality of a poll application by adding user authentication features. Users must register and log in before they can participate in voting.
- **Tools:** Python 3.11, HTML5 CSS Bootstrap,
- **GitHub Repo:** <https://github.com/asandilenkala/Poll-App-and-Authentication>

### **Database Project: [HyperionDev]**

- This program is designed for use by a bookstore clerk to manage an inventory of books. The program allows the clerk to add new books to the database, update book information, delete books from the database, and search the database to find specific books. The program interacts with a database called e-bookstore and manages a table named book.
- **Tools:** Python 3.1, SQLite3
- **GitHub Repo:** <https://github.com/asandilenkala/Database-Project>

### **Mobile Development Design: [UCT]**

- This Android Design App was created by me and my project partner Clifford Silindana, [SLNCLI002@myuct.ac.za](mailto:SLNCLI002@myuct.ac.za)
- **Tools:** Kotlin, XML
- **GitHub Repo:** <https://github.com/asandilenkala/Mobile-Development-Design>

### **Monte Carlo Function Optimization: [UCT]**

- This project demonstrates the parallelization of Monte Carlo Function Optimization using the Java Fork/Join framework and the divide-and-conquer strategy(Multithreading). The primary goal is to optimize a function by leveraging parallel computation to accelerate the Monte Carlo simulation process.
- **Tools:** Java, Makefile
- **GitHub Repo:** <https://github.com/asandilenkala/Parallelizing-Monte-Carlo-Function-Optimisation>

## **Experience**

**Cloud and Software Developer at Hapo Technology.**  
**Aug 2025 - March 2026**

- My Role is a perfect blend of my passion, on both Backend Frontend Development and Deployment.
- Focus on real-world skills and personal growth in a digital economy.
- I had the privilege of leading a project and guiding new interns with technical tools.

**Cloud Associate at CAPACITI.**  
**Aug 2025**

**Feb 2025 -**

- Part of **CAPACITI IT Learnership Programme** in Cape Town.
- Training in cloud infrastructure, deployment, certifications, industry projects.
- Focus on real-world skills and personal growth in a digital economy.

**UCT Information System Tutor, Cape Town**

**2023**

- Tutor for INF1002F & INF1002S Course
- Assisted in Excel & Introduction to Python.
- Responsible for marking Student's Exercises and their Semester Projects

## **EDUCATION**

**Bachelor of Science in Computer Science & Computer Engineering 2019-2023**

**University of Cape Town**

- Computer Science Status: Completed 1<sup>st</sup> year and 2<sup>nd</sup> year 2<sup>nd</sup> semester.
- Computer Engineering Status: Completed 1<sup>st</sup> & 2<sup>nd</sup> year, also completed 3<sup>rd</sup> year 2<sup>nd</sup> semester course but I did not pass 3<sup>rd</sup> year 1<sup>st</sup> semester work.
- **Note: Degree is incomplete, but completing it is part of my goal.**

## **CERTIFICATION**

**HyperionDev Boot Camp**  
**2024**

**March-October**

**Average Grade: 98%**

**Software Engineering Course [NQF Level 5]**

- The Software Engineering Boot Camp focuses on core programming skills, web development, databases, and software design. It covers Python, JavaScript, and SQL, preparing students for real-world software development roles with practical, project-based learning.

## **CAREER GOALS**

- **Becoming a Proficient Software Engineer Launching a Tech Company**
- **Expanding My Knowledge**
- **Build a Strong Portfolio**

- Gain Experience

## **ACCOMPLISHMENTS**

**MyDomain Salt River Residence [Leadership]:  
[2023]**

- Deputy Head of Students
- Disciplinary Committee member
- Entertainment Representative

**Obtaining my Software Engineering Certificate**

## **LANGUAGES**

- IsiXhosa [HL]
- English [FAL]

## **CERTIFICATE**

<https://www.hyperiondev.com/portfolio/244218/>